Equivalence of the Definitions of the Second Fundamental Form

Let $X(u, v)$ be a regular parametrization of a surface $S \subset \mathbb{R}^3$. Denote:

$$X_u = \frac{\partial X}{\partial u}, \quad X_v = \frac{\partial X}{\partial v}$$

and define the unit normal field

$$N = \frac{X_u \times X_v}{\|X_u \times X_v\|}.$$

Definition via Second Derivatives

The coefficients of the second fundamental form are defined as

$$L = \langle X_{uu}, N \rangle, \quad M = \langle X_{uv}, N \rangle, \quad N = \langle X_{vv}, N \rangle.$$

Definition via the Weingarten Map

The Weingarten map (shape operator) is defined by

$$dN_p : T_pS \to T_pS.$$

Its coefficients are defined via:

$$L = -\langle N_u, X_u \rangle, \quad M = -\langle N_u, X_v \rangle = -\langle N_v, X_u \rangle, \quad N = -\langle N_v, X_v \rangle.$$

Goal

Show that:

$$\langle X_{ij}, N \rangle = -\langle N_i, X_j \rangle, \quad i, j \in \{u, v\}.$$

Proof

We use the fundamental orthogonality relation:

$$\langle N, X_u \rangle = 0, \quad \langle N, X_v \rangle = 0.$$

Differentiate $\langle N, X_u \rangle = 0$ with respect to u:

$$\frac{\partial}{\partial u} \langle N, X_u \rangle = \langle N_u, X_u \rangle + \langle N, X_{uu} \rangle = 0.$$

Hence:

$$\langle N, X_{uu} \rangle = -\langle N_u, X_u \rangle,$$

which proves:

$$L = \langle X_{uu}, N \rangle = -\langle N_u, X_u \rangle.$$

Similarly, differentiate $\langle N, X_u \rangle = 0$ with respect to v:

$$\frac{\partial}{\partial v}\langle N, X_u \rangle = \langle N_v, X_u \rangle + \langle N, X_{uv} \rangle = 0,$$

thus:

$$M = \langle X_{uv}, N \rangle = -\langle N_v, X_u \rangle.$$

Likewise, differentiating $\langle N, X_v \rangle = 0$ with respect to u:

$$\langle N_u, X_v \rangle + \langle N, X_{uv} \rangle = 0,$$

so:

$$M = -\langle N_u, X_v \rangle.$$

Finally, differentiate $\langle N, X_v \rangle = 0$ with respect to v:

$$\langle N_v, X_v \rangle + \langle N, X_{vv} \rangle = 0,$$

giving:

$$N = \langle X_{vv}, N \rangle = -\langle N_v, X_v \rangle.$$

Conclusion

We have shown:

$$\langle X_{ij}, N \rangle = -\langle N_i, X_j \rangle,$$

thus the two definitions of the second fundamental form coincide.

$\square$